In [ ]:
```
import pandas as pd
import numpy as np
from scipy import stats
import seaborn as sns
```

Hypothesis testing: Chi-Square Test within the Eniac case study

In this notebook we perform a chi-square test with the data from the Eniac case study, applying a post-hoc correction to perform pairwise tests and find the true winner.

1. State the Null Hypothesis and the Alternative Hypothesis.

Null Hypothesis (H_0): The click-through rate for all versions of the website is equal.

Alternative Hypothesis (H_A): The click-through rate for at least one version of the website differs.

2. Select an appropriate significance level alpha (α).

It was decided that a relatively high alpha was acceptable in this case

In [ ]:
```
alpha = 0.05
```

3. Collect data that is random and independent

The important pieces of information (clicks on each element of interest & visits on each page) are scattered around. Let's collect them.

Importing the csv's

In [ ]:
```
# Element list eniac_a.csv
url = 'https://drive.google.com/file/d/1x4DwDVc7uA2eIMEkmujmv81SfANjm6GC/view?usp=d
path = 'https://drive.google.com/uc?export=download&id='+url.split('/')[-2]
eniac_a_df = pd.read_csv(path)

# Element list eniac_b.csv
url = 'https://drive.google.com/file/d/1Z4vLwetcGbk75cRihAp_rVtQ879vPYhu/view?usp=d
path = 'https://drive.google.com/uc?export=download&id='+url.split('/')[-2]
eniac_b_df = pd.read_csv(path)

# Element list eniac_c.csv
url = 'https://drive.google.com/file/d/1mFk5UkygtUAmwx3KD3y8WRQVkF959YCv/view?usp=d
```

```
path = 'https://drive.google.com/uc?export=download&id='+url.split('/')[-2]
eniac_c_df = pd.read_csv(path)

# Element list eniac_d.csv
url = 'https://drive.google.com/file/d/1e5ldQGnyQIL5UWYe1nDiKReNr1NGNvzg/view?usp=d
path = 'https://drive.google.com/uc?export=download&id='+url.split('/')[-2]
eniac_d_df = pd.read_csv(path)
```

How many clicks did each element get?

In [ ]:
```
pd.set_option('display.max_colwidth', None)
```

In [ ]:
```
eniac_a_df
```

In [ ]:
```
eniac_b_df
```

In [ ]:
```
eniac_c_df
```

In [ ]:
```
eniac_d_df
```

In [ ]:
```
eniac_a_clicks = eniac_a_df.loc[eniac_a_df["Name"]=="SHOP NOW", "No. clicks"].iloc[
eniac_b_clicks = eniac_b_df.loc[eniac_b_df["Name"]=="SHOP NOW", "No. clicks"].iloc[
eniac_c_clicks = eniac_c_df.loc[eniac_c_df["Name"]=="SEE DEALS", "No. clicks"].iloc
eniac_d_clicks = eniac_d_df.loc[eniac_d_df["Name"]=="SEE DEALS", "No. clicks"].iloc
```

In [ ]:
```
eniac_a_df.loc[eniac_a_df["Name"]=="SHOP NOW", "No. clicks"]#.iloc[0]
```

In [ ]:
```
eniac_a_clicks
```

In [ ]:
```
eniac_b_clicks
```

In [ ]:
```
eniac_c_clicks
```

In [ ]:
```
eniac_d_clicks
```

How many visits did each page get (they are in the last column of the second row, we read them manually)?

In [ ]:
```
eniac_a_df.iloc[1, -1]
```

In [ ]:
```
eniac_a_visits = 25326
eniac_b_visits = 24747
eniac_c_visits = 24876
eniac_d_visits = 25233
```

From the above information, we can calculate the number of visitors who didn't click on the button

In [ ]:
```
eniac_a_no_click = eniac_a_visits - eniac_a_clicks
eniac_b_no_click = eniac_b_visits - eniac_b_clicks
```

```
eniac_c_no_click = eniac_c_visits - eniac_c_clicks
eniac_d_no_click = eniac_d_visits - eniac_d_clicks
```

Now we can make a contingency table that shows the clicks and no clicks for each version of the website

In [ ]:
```
clicks = [eniac_a_clicks, eniac_b_clicks, eniac_c_clicks, eniac_d_clicks]
noclicks = [eniac_a_no_click, eniac_b_no_click, eniac_c_no_click, eniac_d_no_click]

observed_results = pd.DataFrame(data = [clicks, noclicks],
                                columns = ["Version_A", "Version_B", "Version_C", "
                                index = ["Click", "No-click"])

observed_results
```

4. Calculate the test result

In [ ]:
```
chisq, pvalue, df, expected = stats.chi2_contingency(observed_results)
```

In [ ]:
```
df
```

In [ ]:
```
expected
```

In [ ]:
```
pvalue
```

5. Interpret the test result

In [ ]:
```
if pvalue > alpha:
  print("Do not reject the null hypothesis")
else:
  print("Reject the null hypothesis")
```

Since the p-value is (much) smaller than alpha, we reject the Null Hypothesis.

> Remember: **If p is low, the Null must go!**

This means that at least one of the four different versions performed significantly better or worse than the others.

But how do we decide who's the winner?

If you feel very brave, read about Post Hoc Tests and find out whether we can declare a clear winner.

> **In brief:** Post hoc tests, also known as post hoc comparisons or pairwise comparisons, are statistical procedures used in conjunction with a Chi-

> Squared test to determine specific differences between groups or conditions after obtaining significant results from the Chi-Squared test. Once the Chi-Squared test reveals a significant difference among the variables being examined, post hoc tests are employed to identify which specific groups or categories are significantly different from each other.

We have 6 possible dual tests to perform:

- Version A - Version B
- Version A - Version C
- Version A - Version D
- Version B - Version C
- Version B - Version D
- Version C - Version D

The level of alpha we selected for the chi-squared test cannot be same for the dual tests. If there was an error of 5% in each of the tests, this would sum up to much more than the 5% total we set for alpha (nearly 26%), so we will need to be much more restrictive in the dual tests. Therefore, we will split the value of alpha equally among the dual tests to be performed.

This is what's called the Bonferroni Correction!

In [ ]:
```
possible_combinations = 6
alpha_post_hoc = alpha / possible_combinations
alpha_post_hoc
```

Before we begin, let's have a look at the click through rates to see the relative success of each version

In [ ]:
```
# click-through rates
eniac_a_ctr = eniac_a_clicks / eniac_a_visits
eniac_b_ctr = eniac_b_clicks / eniac_b_visits
eniac_c_ctr = eniac_c_clicks / eniac_c_visits
eniac_d_ctr = eniac_d_clicks / eniac_d_visits

# display as DataFrame
rates = [eniac_a_ctr, eniac_b_ctr, eniac_c_ctr, eniac_d_ctr]
names = ["Version_A", "Version_B", "Version_C", "Version_D"]

ctr_df = pd.DataFrame({"rates": rates, "names": names})
ctr_df.sort_values("rates", ascending=False)
```

It appears that the two white buttons have achieved the highest level of success. However, to gain a comprehensive understanding of statistical significance, we will conduct a post hoc test across all versions. Nonetheless, our primary emphasis should be on analysing the discrepancy between the white "SEE DEALS" and the white "SHOP NOW" buttons, as well as the disparity between these two and the remaining versions.

In [ ]:
```
observed_results
```

In [ ]:
```
observed_results.columns
```

In [ ]:
```
observed_results.loc[:, ['Version_B', 'Version_D']]
```

In [ ]:
```
# empty dictionary to fill with our results
stat_significant_dict = {
                        "Version_A": [],
                        "Version_B": [],
                        "Version_C": [],
                        "Version_D": []
                         }

# compare each version to each other version
for i in observed_results.columns:
  for j in observed_results.columns:
    # use scipy to find the p-value of each pair
    chisq, pvalue, df, expected = stats.chi2_contingency(observed_results.loc[:, [i
    # boolean: if the p-value is lower than alpha, our result is statistically sign
    stat_significant_dict[i].append(pvalue < alpha_post_hoc)

# create a DataFrame of results
stat_significant_df = pd.DataFrame(stat_significant_dict,
                                   index=observed_results.columns)

# create a heatmap from the DataFrame & red/green colour palette
red_green_palette = sns.diverging_palette(10, 120, n=2, s=70, l=50)
ax = sns.heatmap(stat_significant_df, cmap=red_green_palette)

# Manually specify colorbar labelling after it's been generated
colorbar = ax.collections[0].colorbar
colorbar.set_ticks([0.25, 0.75])
colorbar.set_ticklabels(['False', 'True'])

# Add a title to the heatmap
ax.set_title("Statistical Significance in Pairwise Tests: True or False?", pad=10);
```

In [ ]:
```
stat_significant_dict
```

In [ ]:
```
stat_significant_df
```

Analysing our heatmap, it is evident that the version with the highest click-through rate, `Version_C`, exhibits a statistically significant difference when compared Versions B and D, but not to `Version_A`, which possesses the second-highest click-through rate. As a result, declaring a clear winner based on post hoc tests becomes challenging, therefore we can only say that both `Version_C` and `Version_A` are the winners.

However, if a definitive winner is required, additional steps need to be implemented. This is where we transition from the realm of statistics to the business world. The following actions can help in determining the version to be featured on the website in the future:

- Consider other metrics alongside click-through rate.
- Incorporate qualitative research findings.
- Seek input from subject-matter experts.
- Redesign the experiment and conduct it once more.